

# Rao Faizan

Ph.D. Candidate, Artificial Intelligence — Kyung Hee University, South Korea

[rao.faizan@khu.ac.kr](mailto:rao.faizan@khu.ac.kr) · [faize.rao5445@gmail.com](mailto:faize.rao5445@gmail.com)

[raofaizan.github.io](https://raofaizan.github.io) · [github.com/raofaizan](https://github.com/raofaizan) · [linkedin.com/in/rao-faizan](https://linkedin.com/in/rao-faizan) · [Google Scholar](#) · [ORCID](#)

## Summary

AI researcher and engineer with five years of combined research and teaching experience, currently completing a Ph.D. on multi-agent LLM systems. Builds language and vision models end to end: efficient inference (quantization, knowledge distillation, pruning), vision-language modeling on clinical data, and reasoning-augmented agent pipelines. Delivers government- and industry-funded projects on schedule, with peer-reviewed publications in knowledge graphs and computer vision. Seeking research scientist or machine learning engineer roles in applied LLM and multimodal systems.

## Technical Skills

|                            |                                                                                                                                                                                                                            |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Languages</b>           | Python (primary), C/C++, Java, SQL, JavaScript, HTML/CSS                                                                                                                                                                   |
| <b>ML &amp; DL</b>         | PyTorch, Keras/TensorFlow, Hugging Face Transformers, scikit-learn, NumPy, pandas                                                                                                                                          |
| <b>LLM &amp; Agents</b>    | Fine-tuning (LoRA/PEFT), quantization, knowledge distillation, structured pruning, retrieval-augmented generation, multi-agent orchestration, prompt and tool-use design, neuro-symbolic reasoning and output verification |
| <b>Multimodal &amp; CV</b> | Vision-language models, image captioning and VQA (MIMIC-CXR), person and vehicle re-identification, attention mechanisms, object detection                                                                                 |
| <b>Time Series</b>         | LSTM and sequence models for demand forecasting, scheduling and optimization                                                                                                                                               |
| <b>Tooling</b>             | Git, Linux, Docker, Anaconda, Jupyter, VS Code, Django, Weka, R Studio                                                                                                                                                     |

## Research & Professional Experience

### Graduate Researcher — AI Agents & Language Models

Mar 2024 – Present

*Kyung Hee University, Gyeonggi-do, South Korea*

- Designing a multi-agent LLM system for cardiovascular disease diagnostics and patient health literacy under a Korean government-funded research programme.
- Building lightweight healthcare language models using quantization, knowledge distillation and structured pruning to cut inference cost while preserving diagnostic accuracy.
- Developing vision-language models over the MIMIC chest X-ray corpus for clinical question answering and automated medical report generation.
- Implementing a proof-reasoner component that verifies LLM-generated clinical claims against a symbolic knowledge base, reducing unsupported outputs.
- Co-authoring grant proposals and project deliverables alongside peer-reviewed publications.

### Lecturer — Department of Software Engineering

Oct 2022 – Feb 2024

*University of Central Punjab (UCP), Lahore, Pakistan* | [faculty profile](#)

- Taught seven undergraduate courses across three semesters, including Introduction to Machine Learning, Image Processing, Programming Fundamentals, Databases, Operating Systems and Discrete Structures.
- Supervised undergraduate final-year projects in machine learning and computer vision from proposal through defence; served on the “Become a Globalized University” strategic committee.

### Lecturer — Department of Software Engineering

Sep 2021 – Nov 2022

*University of Lahore (UOL), Lahore, Pakistan*

- Taught Artificial Intelligence, Operating Systems, Human-Computer Interaction, Discrete Structures and Introduction to ICT across the Fall 2021 – Fall 2022 semesters.
- Member of the Final-Year Project committee; organized workshops on L<sup>A</sup>T<sub>E</sub>X document engineering and full-stack web development (MERN, Laravel, React Native).

### Research Assistant — Machine Vision & Intelligent Systems Lab

Feb 2020 – Apr 2021

*National University of Sciences & Technology (NUST), Islamabad, Pakistan*

- Developed IAB-Net, an attention-based deep network for person re-identification across non-overlapping surveillance cameras; owned the full cycle from dataset curation and model design through ablation studies, published and presented at ICODT2 2021.

## Selected Projects

---

### Lightweight Language Model for Healthcare

2024 – Present

Kyung Hee University / Python, PyTorch, Hugging Face

- Neuro-symbolic small language model for medical diagnosis support: health literacy, disease detection, treatment recommendation and medical report generation.
- Owned the model-efficiency workstream (quantization, distillation, pruning) and the vision-language pipeline on MIMIC.

### AI-Based Energy Management Solution — Japan Market

2024 – Present

Kyung Hee University / Python, PyTorch, Keras, LSTM

- Built the Self Demand Response (SDR) component that adjusts household and facility energy use in response to grid supply conditions and price signals.
- Unified heterogeneous consumption datasets and trained LSTM forecasting models driving the SDR policy; designed the battery-storage scheduling algorithm.

## Publications

---

16 citations, h-index 2 — Google Scholar

- [1] **R. Faizan**, S. Lee, and S. T. Kim. “ChestVQATrans: A transformer-based approach for medical VQA in radiology.” *Proc. 8th Artificial Intelligence and Cloud Computing Conference (AICCC)*, pp. 169–173, ACM, Dec. 2025. (First author)
- [2] S. Wazir, **R. Faizan**, and D. Kim. “HistoSeg++: Delving deeper with attention and multiscale feature fusion for biomarker segmentation.” *Proc. 12th International Conference on Biomedical and Bioinformatics Engineering (ICBBE)*, ACM, 2025. doi:10.1145/3794209.3794211
- [3] M. B. Yaseen, **R. Faizan**, M. Waseem, M. Kamran, and N. U. Haq. “Sarcasm detection in news headlines using deep learning with linguistic feature fusion.” *6th International Conference on Innovative Computing (ICIC)*, pp. 1–6, IEEE, Dec. 2025.
- [4] T. D. T. Nguyen, U. U. Rehman, M. Hussain, **R. Faizan**, J. Hussain, S.-H. Bae, J. U. Kim, S. T. Kim, and S. Lee. “Unified link prediction modeling for enhanced knowledge graph completion task.” *Expert Systems with Applications*, vol. 279, 127356, 2025. (SCIE)
- [5] **R. Faizan**, M. M. Fraz, and M. Shahzad. “IAB-Net: Informative and attention-based person re-identification.” *International Conference on Digital Futures and Transformative Technologies (ICoDT2)*, pp. 1–5, IEEE, 2021. (First author)

### Under review

- [U1] C. Bold, **R. Faizan**, C.-H. Hua, S. M. Kang, S. Lee, and S. T. Kim. “Hybrid retrieval-augmented generation for chest-pain differential diagnosis: recall gains and calibration costs.” 2025. Benchmark of 165 PubMed Central case vignettes.

## Education

---

### Ph.D. in Computer Engineering — Artificial Intelligence

Mar 2024 – Present

Kyung Hee University (KHU), Gyeonggi-do, South Korea / CGPA 4.25 / 4.30 (99.33 percentile) / Presidential Scholarship

Coursework: Advanced Computer Vision, AI Advanced Research Project I–II, Future Biomedical Engineering Technology, Blockchain in Medical Technology, Creative Software I–II

### M.S. in Computer Science

Sep 2018 – Apr 2021

National University of Sciences &amp; Technology (NUST), Islamabad, Pakistan

Thesis: Informative Attention-Based Deep Network (IAB-Net) for Person Re-Identification

### B.S. in Computer Science

Oct 2014 – Jun 2018

The Islamia University of Bahawalpur (IUB), Pakistan / 1st position, semesters 2–8 / Bronze Medalist

## Honours, Awards & Certifications

---

- **Presidential Scholarship**, Kyung Hee University — full Ph.D. funding awarded for academic excellence, 2024–2028.
- **Outstanding Performer**, Federal Employees Benevolent & Group Insurance Funds (FEBGIF), Pakistan, 2021.
- **Bronze Medalist and 1st position (semesters 2–8)**, The Islamia University of Bahawalpur, 2018.
- **Certifications**: MIT CITI Data & Specimens Only Research (2024) · IBM Data Analysis and Machine Learning with Python (2023) · PIAIC Certified Cloud Native Developer (2023) · Harvard Launching Breakthrough Technologies (2019).